

Industrial Energy Demand Analysis Report

Bismillah Foundry and Light Casting Ltd
Bright PET Washing Line Industries
BFL Gate, Rajarhat, Chuknagar Road, Jashore



Submitted by	Supervised by
- Abdullah Shadek Fahim (191131) - Md. Shakil Hossain (201136) - Irin Akter Mim (191118) - Nure Modina Sathi (191122)	Dr. Engr. Imran Khan Associate Professor Department of Electrical and Electronic Engineering Jashore University of Science and Technology

Submission Date: 01 June, 2025



Department of Electrical and Electronic Engineering
Jashore University of Science and Technology

1st Floor, Sir Jagadish Chandra Bose Academic Building
Phone: 02-42142031, PABX: 220
chairman.eee@just.edu.bd

Disclaimer

This report presents the findings of an energy audit conducted by a group of 4th year, 1st Semester students of the Department of Electrical and Electronic Engineering (EEE), Jashore University of Science and Technology as part of the Energy Management & Efficiency (EEE-4121) course under the supervision of Dr. Engr. Imran Khan (Associate Professor, Dept of EEE, JUST). As students, the auditors are not certified energy auditors. This exercise was carried out for academic purposes, using standard auditing methods and available data. While every effort was made to ensure accuracy, the findings and recommendations should be reviewed by qualified professionals before implementation. The instructors for their guidance and the facility team for their cooperation and support are gratefully acknowledged.

Acknowledgement

We, the audit team, extend our sincere gratitude to **Md. Soaib Hossain**, Managing Director of Bismillah Foundry and Light Casting, Bright PET Washing Line Industries, for graciously providing us with the invaluable opportunity to visit the industry. His extensive cooperation and the wealth of information he shared were instrumental to the success of this energy audit.

Our profound thanks also go to our esteemed course teacher, **Dr. Engr. Imran Khan**, Associate Professor, Department of Electrical and Electronic Engineering (EEE), Jashore University of Science and Technology. His continuous encouragement and guidance were pivotal in pushing us to gain these invaluable real-life field-level experiences, significantly enriching our practical understanding of energy management and auditing.

Table of Contents

Acknowledgement	3
Executive Summary	5
Summary of Baseline Energy and Resource Consumption	5
1.0 Facility Overview	5
1.1 General Plant Details and Descriptions	5
1.2 Energy Audit Team Details.....	6
1.3 Scope, Methodology, and Instruments Used.....	6
1.4 Major Energy Use Areas.....	6
2.0 Production Process Description	7
2.1 Brief Description of the Manufacturing Process.....	7
3.0 Energy and Utility System Description	9
3.1 List of Utilities	9
3.2 Brief Description of Each Utility	9
4.0 Energy Use Analysis and Benchmarking	11
4.1 Overall Energy Consumption.....	11
4.2 Key Observations from Electricity Bill Analysis:	12
4.3 Power Factor Issue and Surcharge Discrepancy	12
5. Lighting and Waste Management Analysis	14
5.1 Lighting Analysis.....	14
5.2 Waste Management System	15
5.3 Sound and Noise Level	15
6.0 Energy Conservation Measures (ECMs) and Recommendations.....	15
6.1 Critical Power Factor Correction	16
6.2 Establish Comprehensive Energy Monitoring and Management System.....	16
6.3 Lighting System Upgrades and Optimization	16
6.4 Motor and Equipment Optimization	17
6.5 Water System Efficiency	17
6.6 HVAC System Improvements	17
6.7 Renewable Energy Integration.....	18
6.8 Operational and Maintenance Best Practices	18
7.0 Conclusion	18
References	19

Executive Summary

This energy audit was meticulously conducted at Bismillah Foundry and Light Casting, Bright PET Washing Line Industries, situated at BFL Gate, Rajarhat, Chuknagar Road, Jashore. The audit, performed on May 11, 2025, aimed to comprehensively assess the existing energy consumption patterns and identify tangible opportunities for both cost reduction and enhancement of operational efficiency within the facility. The industry, a multifaceted manufacturing entity, engages in Engineering and Mechanical, Water Supply and Irrigation, Recycling and Petrochemical, and Export and International Trade, producing a diverse range of cast iron components and recycled PET flakes.

A critical finding from the analysis of the past year's electricity bills is the alarmingly low power factor, which consistently incurs substantial penalties. The average monthly electricity consumption, encompassing both Off-Peak and Peak periods, stands at approximately 3208 kWh, with an average monthly peak demand of 625 kWh. The cumulative annual electricity expenditure amounts to a significant 761,374 BDT. This report underscores the imperative need for immediate and sustained power factor correction initiatives, alongside a strategic optimization of overall energy usage, to mitigate financial losses and foster a more energy-efficient operational environment.

Summary of Baseline Energy and Resource Consumption

- **Average Monthly Electricity Usage (Off-Peak & Peak):** Approximately 3208 kWh
- **Average Monthly Peak Demand:** Approximately 625 kWh
- **Annual Electricity Cost:** 761,374 BDT
- **Water Usage:** Approximately 32,000 Liters/day
- **Primary Energy Source:** Grid Electricity and Diesel Generator
- **Average Measured Power Factor:** 0.52 (ranging from 0.33 to 0.96 over the year)

Energy Saving Options at a Glance and Recommendations

The audit has identified several key areas for energy conservation, categorized by their potential impact and implementation complexity. These include critical power factor improvement, comprehensive energy monitoring, lighting system upgrades, optimization of motors and pumps, and the activation of existing renewable energy infrastructure. Detailed recommendations are elaborated in Section 6.0.

1.0 Facility Overview

1.1 General Plant Details and Descriptions

Bismillah Foundry and Light Casting, Bright PET Washing Line Industries, is a prominent manufacturing facility located at BFL Gate, Rajarhat, Chuknagar Road, Jashore. The industry's operations span across diverse sectors including engineering, mechanical, water supply, irrigation, recycling, and petrochemical, with a significant involvement in export and international trade. The facility operates consistently from 8 AM to 10 PM, with potential variations on Fridays.

The core manufacturing activities revolve around the production of various cast iron components such as hand pumps, centrifugal pumps, oil mill parts, rice milling machine components, bladed drums, gear housings, bus brake drums, hoppers, vehicle components, engine cylinder heads, and brake systems. Additionally, the industry is a key producer of recycled PET flakes in brown, clear, and green varieties. The total operational area of the facility extends over 73,950 sq ft, primarily consisting of a single level with cement flooring and a tin rooftop. The plant is divided into five (5) distinct compartments. The surrounding environment features a main road on one side and agricultural land interspersed with water

bodies like ponds or small lakes on the other. The workforce comprises over 100 employees, supported by 4 executives and one engineer from BUET. The production capacity is notable, with 100 tons/month for cast iron and 200 tons/month for PET flakes.

1.2 Energy Audit Team Details

The energy audit was conducted by a dedicated team of undergraduate students from the Department of Electrical and Electronic Engineering (EEE), Jashore University of Science and Technology (JUST), under the expert supervision of Dr. Engr. Imran Khan, Associate Professor, Dept. of EEE, JUST. The audit team members included:

- Abdullah Shadek Fahim (191131)
- Md. Shakil Hossain (201136)
- Irin Akter Mim (191118)
- Nure Modina Sathi (191122)

The primary point of contact and cooperation from the industry was Md. Soaib Hossain, Managing Director, who can be reached at +8801714082862 or soaib.hossain.1980@gmail.com.

1.3 Scope, Methodology, and Instruments Used

The audit's overarching objective was to conduct a thorough assessment of the industry's energy consumption patterns and to pinpoint viable opportunities for enhancing energy efficiency, thereby leading to reduced operational costs. The audit was carried out on May 11, 2025, from 9 AM to 1:30 PM. The weather conditions during the audit were recorded as 38° C, sunny, with 45% humidity and a wind speed of 10 KM/h, and a perceived temperature of 30%.

Key instruments utilized during the audit included a Tachometer for rotational measurements, a Lux meter for lighting assessments, a 15 ft measurement tape for spatial dimensions, and a Sound meter App for noise level measurements.



Fig 1. Tachometer



Fig 2. Luxmeter



Fig 3. Measurement Tap

1.4 Major Energy Use Areas

The primary energy-consuming areas within the facility are predominantly associated with the operational machinery for both PET flakes processing and iron casting. Key equipment identified as major energy consumers include:

- PET Flakes Washing Machine
- Crusher machine
- Washing machine
- Dryers
- Conveyors
- Drilling machines

Each of these machines is driven by motors, with individual power consumption ranging from 5 kW to 15 kW.

Other significant equipment observed during the audit that contribute to the overall energy demand include:

- PET Bottle Shredder
- PET Washing Line
- Metal Separator
- PET Flake Dryer
- Industrial Steam Boiler
- Recycling Furnace
- Hydraulic Press
- Industrial Shearing Machine
- Grinder machine
- Welding Machine
- Industrial Drilling Machine
- Iron Cutter
- Metal Casting Machine
- Lathe Machine.

2.0 Production Process Description

2.1 Brief Description of the Manufacturing Process

The industrial operation at Bismillah Foundry and Light Casting, Bright PET Washing Line Industries, involves distinct processes for both PET flakes [1] recycling and iron casting. The PET flakes production follows a sequential process:

1. **Collection and Sorting:** Raw PET bottles are collected and sorted to remove impurities and separate by color.
2. **Metal Separation:** Any metallic contaminants are removed.
3. **Shredding:** The sorted PET bottles are shredded into smaller pieces.
4. **Hot Water Washing:** Shredded PET is washed with hot water to remove labels and adhesives.
5. **Caustic Washing:** A caustic solution is used for further cleaning and decontamination.
6. **Drying:** The washed flakes are dried to remove moisture.
7. **Flake Sorting:** Further sorting ensures quality and purity of flakes.
8. **Hot Air Drying:** Additional drying using hot air.
9. **Decontamination:** Final decontamination steps.
10. **Packaging:** The finished PET flakes Flakes (Brown, Clear and Green) are packaged for distribution.

The iron casting process involves melting raw iron in a furnace and then pouring it into molds to create various cast iron products. Which later turns into various products like 6 No. Hand Pump, Centrifugal Pump, Oil Mill, Rice Milling Machine, Parts of Jute Processing Mill, Bladed Drum, Gear Housing, Bus Brake Drum, Hopper, Vehicle Components, Engine Cylinder Head, Brake Systems,. The facility also utilizes automated systems for the PET flakes machine and waste management, indicating some level of modern integration in these areas.



Fig 4. PET Flakes Washing Machine



Fig 5. Cast iron manufacturing product compartment

PET Flakes Washing Line (Zhangjiagang Sevenstars Machinery co., Ltd)

- Processing capacity: 300~500 kg/h
- Finished product: blue and white bottle flakes, green bottle flakes, pure white PET bottle flakes; humidity below 1%, impurities below 100ppm
- **Equipment Power Consumption:**
 - Conveyor 1: 1.5 kW
 - Bottle Labeling machine: 15 kW
 - Conveyor 2: 2.2 kW
 - PET plastic crusher: 22 kW
 - Washing tank: 4 kW
 - PET friction washing machine: 4 kW
 - Dehydration machine: 11 kW
 - Total PET Flakes Machines Power: 65 kW

Other Key Equipment:

- Steam Boiler (Small) [2]: 5 kW
- Recycling Furnace (Medium) [3]: 10 kW
- Hydraulic Press [4]: 5 kW
- Grinder Machine [5]: 1 kW
- Welding Machine [6]: 3 kW
- Drilling Machine [7]: 1 kW
- Iron Cutter [8]: 2 kW
- Lathe Machine: 1.5 kW



Fig 6. Grinding Machine



Fig 7. Drilling Machine



Fig 8. Lathe Machine

3.0 Energy and Utility System Description

3.1 List of Utilities

The primary utilities consumed by the industry are electricity and water.

3.2 Brief Description of Each Utility

3.2.1 Electrical System

The facility's electrical supply is drawn from the national grid, supplemented by an oil-based generator for backup. The incoming voltage from the grid was measured at 13.77 kV.

Transformer Details:

- **Type:** MEL/E/16
- **Power Rating:** 315 kVA
- **High Voltage (HV):** 11,000 V
- **Low Voltage (LV):** 415 V / 240 V
- **HV Current:** 16.53 Amps
- **LV Current:** 438.24 Amps
- **Cooling Type:** ONAN (Oil Natural Air Natural)
- **Insulation Level:** LI 75 AC 28/AC 2.5
- **Impedance at 75°C:** 4.5%
- **Oil Weight:** 365 kg
- **Total Weight:** 1590 kg
- **Tap Changer Positions:** 102.5%, 100%, 97.5%, 95%, 92.5%
- **Connection:** DYn11
- **Voltage (Measured):** 13.77 kV
- **Current (Measured):** 0.96 A
- **Power Factor (Estimated):** ~0.9 (Rated).
- **Year of Manufacture:** 2011



Fig 9. Step down Transformer



Fig 10. Transformer ratings



Fig 11. Panel of the Transformer

Distribution and Control:

- **Wiring Type:** A mix of open and conduit-based wiring is observed throughout the facility. No electrical leakage was detected during the audit.
- **Circuit Breakers:** The electrical panels primarily utilize Molded Case Circuit Breakers (MCCBs), rated for protection at 32A, 380V [9].
- **Switchboard Condition:** While the exterior of the switchboards appeared dusty, the internal components were found to be in good condition and adequately protected within their enclosures.

Power Factor Correction (PFC) System:

- **Device:** Mikro PFR120 (Power Factor Regulator) [10]
- **Voltage:** 415/440V
- **Control:** Automatic
- **Steps:** 12 Steps, 50 Hz system
- **Auxiliary Voltage:** 380~415 V AC
- **Capacitor Ratings:** Includes 7.5 kVAR and 15 kVAR capacitors (440V, 50 Hz).
- **Current Power Factor** (from May electricity bill): 0.42



Fig 12. Power Factor Improvement Device



Fig 13. Panel of PFI Device

Backup Power:

- An oil-based generator is available for backup electricity supply, though it was not observed, and no operational data was collected during the audit.

Energy Management:

- Currently, there is no formal energy management system in place, nor is there a designated responsible person for continuous energy consumption monitoring.

3.2.2 Water & Pumps

The facility relies on underground water sources for its operations.

- **Daily Usage:** Approximately 32,000 Liters per day (an approximated value based on available data).
- **Pumps:** The water supply system includes 3 submersible pumps and 4 tubewells. Specific pump ratings could not be taken as they were submerged.
- **Pump Operating Hours:** Pumps operate for approximately 8 hours per day.
- **Water Reservoirs:** Three water reservoirs are present: one large reservoir spanning 1764 sq ft, and two smaller ones with capacities ranging from 500-1000 Liters.
- **Rainwater Harvesting:** No rainwater harvesting system is currently implemented.
- **Drinking Water:** Tubewell water is used for drinking.
- **Pump Control:** A responsible person is assigned for pump control, but there is no automated system to manage pump operations.

3.2.3 HVAC System

The industrial facility primarily relies on natural ventilation for its heating, ventilation, and air conditioning (HVAC) needs, operating continuously for 24 hours a day. Dedicated mechanical exhaust fans are notably absent from the production areas, indicating a reliance on passive airflow for air exchange.

Current HVAC and Cooling Equipment Inventory:

- **Office Room:** Equipped with one (1) air conditioning (AC) unit and two (2) fans.
 - **AC Unit:** 1-ton capacity, rated at 12,000 BTU, with a power consumption of 1.5 kW. It operates for approximately 6-7 hours per day.
 - **Fans:** Each fan is rated at 50 W and operates for approximately 12 hours per day.
- **Compartment 1:** Contains four (4) fans, however, two (2) of these fans were observed to be non-functional during the audit. The spacing between fans and AC units varies (10 ft, 6 ft, 3 ft), which might affect air distribution effectiveness.

4.0 Energy Use Analysis and Benchmarking

4.1 Overall Energy Consumption

The annual energy consumption profile of the industry, as derived from the electricity bill from february 2024 to March 2025, indicates an average monthly electricity usage of 3250 kWh and an average monthly peak demand of 625 kWh over a one-year period. The total annual expenditure on electricity amounts to 761,374 BDT. Detailed analysis mentioned in figure 14. [11]

Billing Month	Date of Previous Rating	Date of Current Rating	Previous Off Peak KWh	Current Off Peak KWh	Previous Peak KWh	Current Peak KWh	Previous Off Peak KVarh	Current Off Peak KVarh	Previous Peak KVarh	Current Peak KVarh	Off Peak Difference (KWh)	Peak Difference (KWh)	Off Peak Difference (KVarh)	Peak Difference (KVarh)	OMF
Mar-25	28-02-25	31-03-25	877	882	181	182	169	171	542	553	5	1	2	11	500
Feb-25	31-01-25	28-02-25	872	877	180	181	165	169	532	542	5	1	4	10	500
Jan-25	30-12-25	30-01-25	867	872	179	180	162	165	520	532	5	1	3	12	500
Dec-24	30-11-24	31-12-24	860	867	178	179	158	162	508	520	7	1	4	12	500
Nov-24	31-10-24	30-11-24	856	860	176	178	154	158	495	508	4	2	4	13	500
Oct-24	30-09-24	31-10-24	851	856	174	176	152	154	487	495	5	2	2	8	500
Sep-24	31-08-24	30-09-24	846	851	173	174	149	152	478	487	5	1	3	9	500
Aug-24	31-07-24	31-08-24	842	846	172	173	147	149	471	478	4	1	2	7	500
Jul-24	30-06-24	31-07-24	836	842	171	172	146	147	466	471	6	1	1	5	500
Jun-24	31-05-24	30-06-24	832	836	170	171	144	146	460	466	4	1	2	6	500
May-24	30-04-24	31-05-24	826	832	168	170	143	144	457	460	6	2	1	3	500
Apr-24	31-03-24	30-04-24	820	826	167	168	143	143	455	457	6	1	0	2	500

Billing Month	Off Peak Kwh Unit Consumption	Peak KWh Unit Consumption	Power Factor Calculated	Given PF	PF Penalty Calculated	PF Surcharge	Off Peak Total (tk)	Peak Total (tk)	Fixed Demand Charge (Tk)	VAT (5%)	Total Bill (Tk)	RoundUP
Mar-25	2500	500	0.419058	0.42	22539.72	4677.75	24375	6810	22500	2918.137	61280.88	61281
Feb-25	2500	500	0.393919	0.39	23606.93	4677.75	24375	6810	22500	2918.137	61280.88	61281
Jan-25	2500	500	0.371390	0.37	24568.40	4911.75	24375	6810	22500	2929.837	61526.58	61527
Dec-24	3500	500	0.447213	0.45	25232.49	6140.25	34125	6810	22500	3478.762	73054.01	73055
Nov-24	2000	1000	0.332820	0.33	27147.92	4968	19500	13620	22500	3029.4	63617.4	63618
Oct-24	2500	1000	0.573462	0.57	18018.65	5699.25	24375	13620	22500	3309.712	69503.96	69504
Sep-24	2500	500	0.447213	0.45	21344.46	4677.75	24375	6810	22500	2918.137	61280.88	61281
Aug-24	2000	500	0.485642	0.49	17917.62	3946.5	19500	6810	22500	2637.825	55394.32	55395
Jul-24	3000	500	0.759256	0.76	8833.077	5138.55	29250	6810	22500	3184.927	66883.47	66884
Jun-24	2000	500	0.529998	0.53	16206.10	3946.5	19500	6810	22500	2637.825	55394.32	55395
May-24	3000	1000	0.894427	0.89	2864.845	1929.15	29250	13620	22500	3364.957	70664.10	70665
Apr-24	3000	500	0.961523	0.96	-531.4383	0	29250	6810	22500	2928	61488	61488
Total in Year											761368.8	761374

Fig 14. Yearly Energy analysis report table from electricity bill

4.2 Key Observations from Electricity Bill Analysis:

- **Total Energy Consumption:** Over the 12-month period, the industry consumed a total of 38,500 kWh of electricity. This averages out to approximately 3208 kWh per month. The consumption pattern shows some variability, with higher usage months (e.g., December 2024, May 2024 at 4000 kWh) and lower usage months (e.g., August 2024, June 2024 at 2500 kWh). These fluctuations could be correlated with production volumes or seasonal operational demands.
- **Off-Peak vs. Peak Usage Distribution:** While off-peak consumption generally constitutes the larger portion of the total energy usage, there are months where peak consumption is relatively higher (e.g., November 2024 and May 2024, both with 1000 kWh peak usage). Understanding the drivers behind these peak demands is crucial for demand-side management.
- **Alarmingly Low Power Factor (PF):** The most critical finding from the electricity bill analysis is the consistently and alarmingly low power factor. The annual average power factor is a mere 0.52. Many months exhibit extremely poor power factors, plummeting to 0.33 in November 2024 and 0.37 in January 2025. This indicates a severe inefficiency in the electrical system, where a significant portion of the total current drawn is reactive power, which does no useful work but still loads the electrical infrastructure.
- **Fixed Demand Charge:** A fixed demand charge of 22,500 BDT is consistently applied every month, totaling 270,000 BDT annually [12]. This charge is levied based on the contracted maximum demand and is independent of actual energy consumption.
- **Total Annual Electricity Cost:** The cumulative electricity bill for the analyzed year amounts to approximately 761,368.86 BDT, a significant operational expense for the industry.

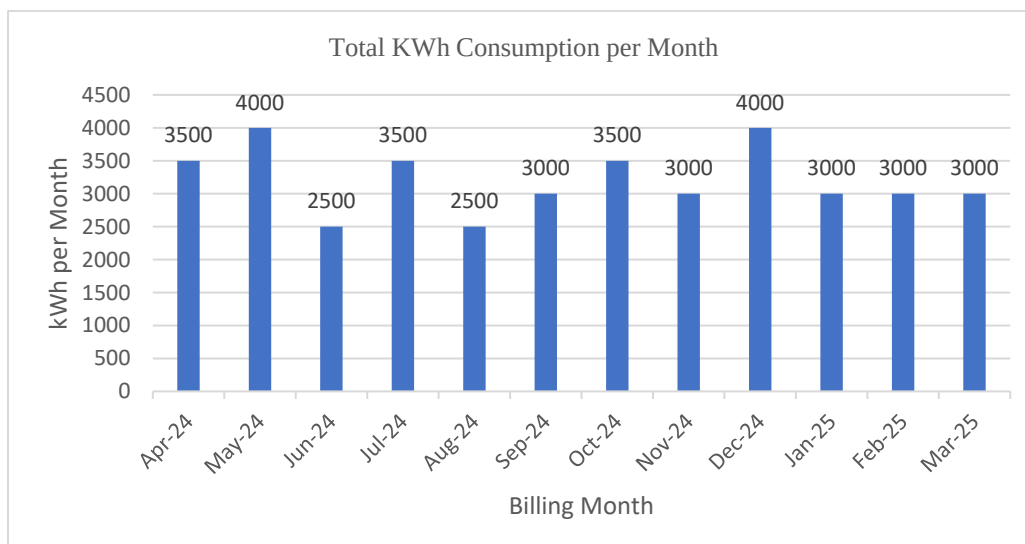


Fig 15. Total KWh Consumption per Month over the year 2024-25

4.3 Power Factor Issue and Surcharge Discrepancy

A critical aspect of the facility's electrical efficiency, and a significant contributor to its operational costs, is the power factor (PF). The audit has revealed consistently low power factor values, ranging from an exceptionally low 0.33 to a high of 0.96 over the past year, with an annual average of 0.52 as highlighted in Section 4.1. This fluctuating and often poor power factor necessitates a deeper analysis.

$$\text{Power Factor [13]} = \frac{\text{kwh}}{\sqrt{(\text{Kwh})^2 + (\text{KVarh})^2}} \dots\dots\dots \text{eq 1}$$

Underlying Causes of Low Power Factor: The industry possesses a Power Factor Improvement (PFI) unit equipped with approximately 20 capacitors, each rated at 25 kVAr, summing up to a substantial total compensation capacity of 500 kVAr. This capacity, when compared to the facility's approved load of 250 kW and a remarkably low recorded maximum monthly load of only 15 kW, points towards a significant imbalance. This imbalance strongly suggests that the system may be experiencing overcompensation with reactive power relative to the active power being consumed. While typically a low power factor indicates a high inductive load (under-compensation), an extremely low actual load combined with a large, potentially active PFI system can lead to the PFI unit injecting more reactive power than required, pushing the power factor into the leading (capacitive) region or causing oscillations around the target, resulting in inefficient or unstable PF correction.

Other contributing factors to the observed low power factor could include:

- **Faulty or Disconnected Capacitors:** Individual capacitors within the PFI unit may be faulty, degraded, or disconnected, preventing the system from delivering its full compensation capacity.
- **Lack of an Effective Automatic Power Factor Correction (APFC) System:** While a Mikro PFR120 (Power Factor Regulator) is present, the consistently low PF values suggest that its automatic control may not be functioning optimally. An APFC system dynamically switches capacitor banks in and out to match the varying reactive power demand of the load. If this system is not operating correctly, capacitors may remain switched on even when not needed, leading to over-compensation during low-load periods, or fail to switch on when required during high inductive loads.
- **Highly Inductive Equipment and Low Active Power Usage:** The nature of the industry involves heavy machinery and motors, which inherently draw significant reactive power. If these inductive loads are operating intermittently or are oversized for the actual production demands, they will draw excessive reactive power relative to the low active power usage, pushing the power factor down. The large discrepancy between the approved load (250 kW) and the actual maximum monthly load (15 kW) means that even relatively small inductive loads can disproportionately impact the power factor.

Power Factor Surcharge Discrepancy Analysis: The Bangladesh Energy Regulatory Commission (BERC) tariff guidelines clearly state that a power factor surcharge is applied when an industrial or commercial consumer's monthly average power factor at the supply point falls below 0.95. Specifically, if the average PF lies between 0.75 and 0.95, a surcharge is imposed at a rate of 0.75% of the energy charge (kWh) for every 0.01 decrease in PF below 0.95. If the PF falls below 0.75, the penalty is capped, and the surcharge is calculated up to the limit of 0.75 PF only [12]. The formula for calculating the surcharge, as provided, is:

$$\text{PF Surcharge} = \frac{(0.95 - \text{Actual PF})}{0.01} \times 0.75\% \times \text{Energy Charge} \dots \dots \dots \text{eq 2}$$

The consistent discrepancy shows in detailed analysis table (fig)for each month suggests that the formula or the base for calculation (Energy Charge) used in the provided BERC guidelines may differ from what the MT 3 customer who have PFI installed, or there might be additional factors not accounted for in the provided formula.

5. Lighting and Waste Management Analysis

5.1 Lighting Analysis

The lighting infrastructure varies across the facility's compartments, with a reliance on both natural daylight and artificial lighting.

- **Compartment 1 (Drilling, Lathe machine etc.)**
 - **Room Size:** 45 ft x 20 ft and mounting height is 10 ft
 - **Lights:** 4 lights, with 1 non-functional.
 - **Type/Rating:** Outdoor/Indoor LED Bulb 15W.
 - **Lux Levels (Measured):** Normal daylight 439 Lumen; With artificial light at day 1077 Lumen.
 - **Energy Consumption per Light:** 20 W at full capacity.
 - **Observation:** The presence of a non-functional light indicates a need for immediate maintenance.
 - **Calculation:** Room size = 45 ft × 20 ft = 900 ft² = 83.61 m²
For Outdoor shaded work areas: 100–200 lux (Standard) [14]
Total Lumens=(0.8×0.6)/(100×83.61)=0.488,361≈17,419lumens
A 20W LED floodlight typically gives 1800–2000 lumens
Let's assume each 20W floodlight gives 1900 lumens.
Number of Lights=17,419/1900≈9.16⇒9 to 10 floodlights needs to be installed. The lights should be 2 rows × 5 columns, spaced evenly with 7.5 ft lengthwise and 6.67 ft widthwise.
- **Compartment 2 (Shaper, furnace)**
 - **Room Size:** 100 ft x 60 ft and mounting height 18 ft
 - **Lights:** 6 lights, all functional.
 - **Type/Rating:** Industrial LED Flood light 250 W.
 - **Lux Levels (Measured):** Normal daylight 196 Lumen; With artificial light at day 779 Lumen.
 - **Energy Consumption per Light:** 250 W at full capacity.
 - **Observation:** The use of extremely high-wattage industrial floodlights warrants a detailed review for potential energy savings through more efficient alternatives.
 - **Calculation:** Room size = 100 ft × 60 ft = 6000 ft² = 557.4 m²
For Outdoor shaded work areas: 100–200 lux (Standard)
Total Lumens = (100 × 557.4)/(0.8 × 0.6) = 55,740/0.48 ≈ 116,125 lumens
A 250W LED floodlight typically gives 100–130 lumens/W [14]
Let's assume each 250W floodlight gives 110 lumens/W ⇒ 250 × 110 = 27,500 lumens
Number of Lights = 116,125 / 27,500 ≈ 4.22 ⇒ 4 to 5 floodlights need to be installed.
The lights should be 1 row × 5 columns, spaced evenly with 20 ft lengthwise and 12 ft widthwise.
- **Compartment 3 (PET Flakes)**
 - **Room Size:** 70 ft x 40 ft, Mounting height 15 ft
 - **Lights:** 6 lights, with 2 non-functional.

- **Type/Rating:** 4 x Industrial LED Flood light 50 W, 2 x Outdoor/Indoor LED Bulb 15W.
- **Lux Levels (Measured):** For 15W lights: Normal daylight 72 Lm; With artificial light at day 126 Lm. For 1500W lights: Normal daylight 138 Lm; With artificial light at day 286 Lm.
- **Observation:** Similar to Compartment 2, the presence of high-wattage floodlights and non-functional lights indicates areas for significant improvement.
- **Calculation:** Room size = 70 ft × 40 ft = 2800 ft² = 260.13 m²
 For Outdoor shaded work areas: 100–200 lux (Standard)
 Total Lumens = $(100 \times 260.13) / (0.8 \times 0.6) = 26,013 / 0.48 \approx 54,194$ lumens
 A 50W LED floodlight typically gives 100–130 lumens/W
 Let's assume each 50W floodlight gives 110 lumens/W $\Rightarrow 50 \times 110 = 5,500$ lumens
 Number of Lights = $54,194 / 5,500 \approx 9.85 \Rightarrow 10$ floodlights need to be installed.
 The lights should be 2 rows × 5 columns, spaced evenly with 14 ft lengthwise and 20 ft widthwise.

Common Lighting Observations:

Daylight control is utilized in all compartments, suggesting an awareness of natural light benefits. However, the wall colors (brown in Compartments 1 & 2, close to gray in Compartment 3) and low reflection indices (0.10 or 0.21) suggest that natural light is not being optimally reflected and utilized, potentially leading to higher reliance on artificial lighting.

5.2 Waste Management System

The industrial facility employs a waste management system that integrates waste disposal with energy recovery. The primary method involves the combustion of generated waste materials, predominantly plastics and other mixed materials, within a dedicated boiler. A notable aspect of this system is the subsequent utilization of the heat generated from this combustion process to fuel the furnace used for making cast iron and other iron materials. This approach represents a form of waste-to-energy conversion, aiming to offset a portion of the primary fuel requirements for the casting process. The facility does not possess an Effluent Treatment Plant.



Fig 16. Waste burning boiler

5.3 Sound and Noise Level

The measured sound level through Sound Meter App in the facility found 83 dB. Which indicates moderate noise level in the industry. Though, while measured all the motors and machines were not running. The noise level may increase when all the machines actively working which can be health issue for the workers.

6.0 Energy Conservation Measures (ECMs) and Recommendations

Based on the comprehensive audit findings and the detailed electricity bill analysis, the following energy conservation measures (ECMs) are recommended to improve energy efficiency, reduce

operational costs, and enhance the overall sustainability of Bismillah Foundry and Light Casting, Bright PET Washing Line Industries.

6.1 Critical Power Factor Correction

- **Observation:** The most pressing issue identified is the consistently low power factor (averaging 0.52), leading to annual penalties exceeding 200,000 BDT. While a Mikro PFR120 (Power Factor Regulator) is installed, its effectiveness is clearly compromised.
- **Recommendation:**
 - **Immediate Investigation and Optimization:** Conduct an urgent and thorough investigation into the existing Mikro PFR120 system. This should include:
 - Verification of capacitor bank health and functionality.
 - Assessment of the automatic control logic and settings.
 - Recalculation of required reactive power compensation based on current and projected loads.
 - Ensuring proper connection and operation of all capacitor steps.
 - **Capacity Enhancement:** If the existing system is undersized or faulty, consider upgrading or adding new capacitor banks to achieve and consistently maintain a power factor of at least 0.95. This will eliminate or drastically reduce the power factor penalties, providing a rapid return on investment.
 - **Regular Monitoring:** Implement a system for continuous monitoring of the power factor to ensure the correction system is operating optimally.

6.2 Establish Comprehensive Energy Monitoring and Management System

- **Observation:** The absence of a formal energy management system or a designated person for energy monitoring limits the ability to track consumption, identify inefficiencies, and measure the impact of energy-saving initiatives.
- **Recommendation:**
 - **Implement an Energy Monitoring System:** Install sub-meters on major energy-consuming equipment (e.g., PET flakes line, casting furnace, large motors) to gather granular data on energy consumption. This will allow for precise identification of energy hogs and operational inefficiencies.
 - **Designate an Energy Coordinator:** Appoint a dedicated individual or a small team responsible for collecting, analyzing, and reporting energy data. This person/team should also oversee the implementation and monitoring of energy conservation measures.
 - **Develop an Energy Management Plan:** Establish a structured energy management plan, potentially aligning with principles of ISO 50001, to set energy targets, track progress, and foster a culture of energy efficiency within the organization.

6.3 Lighting System Upgrades and Optimization

- **Observation:** Several non-functional lights were observed, and high-wattage (250W) industrial LED floodlights are used in large compartments, potentially leading to over and low illumination in certain areas, especially where daylight is available. The low reflection index of wall colors also hinders effective light utilization.
- **Recommendation:**
 - **Repair/Replace Non-Functional Lights:** Immediately repair or replace all non-functional lights to ensure adequate illumination and safety.
 - **LED Retrofit and Right-Sizing:** Conduct a detailed lighting audit to determine the optimal lux levels required for each work area. Replace oversized floodlights with more energy-

efficient LED alternatives that provide appropriate illumination without excessive power consumption.

- **Optimize Daylight Utilization:** Repaint walls and ceilings in lighter, high-reflectance colors (e.g., white or off-white) to maximize the reflection and diffusion of natural light, thereby reducing the reliance on artificial lighting during daytime hours.
- **Implement Lighting Controls:** Install occupancy sensors in intermittently used areas (e.g., storage rooms, restrooms) and timer controls in areas with fixed operating schedules to ensure lights are only on when needed.

6.4 Motor and Equipment Optimization

- **Observation:** The audit identified numerous motors across various equipment without detailed operational efficiency data (e.g., measured RPM vs. rated RPM). Two out of four fans in Compartment 1 were non-functional.
- **Recommendation:**
 - **Detailed Motor Efficiency Assessment:** Conduct a comprehensive assessment of all major motors within the facility. This should include measuring actual operating parameters (current, voltage, power factor, RPM, load) and comparing them against rated specifications to identify oversized or inefficient motors.
 - **Regular Maintenance:** Establish a robust preventive maintenance schedule for all motors, pumps, and other rotating equipment. This includes lubrication, bearing checks, and alignment to ensure optimal operating efficiency and prevent premature failures.
 - **Repair/Replace Non-Functional Fans:** Immediately repair or replace the two non-functional fans in Compartment 1 to improve air circulation and worker comfort.

6.5 Water System Efficiency

- **Observation:** The facility consumes approximately 32,000 Liters of underground water daily, with pumps operating 8 hours a day. There is no automated system for pump control, and pump ratings were not taken.
- **Recommendation:**
 - **Pump Efficiency Assessment:** Conduct an assessment of pump efficiency by measuring flow rates, pressures, and power consumption. Identify opportunities for pump optimization, such as replacing oversized or inefficient pumps with appropriately sized, high-efficiency models.
 - **Automated Pump Control:** Implement an automated system for pump control based on water level sensors in reservoirs or demand patterns. This can prevent unnecessary pump operation and reduce energy consumption.
 - **Leak Detection and Repair:** Regularly inspect the water distribution system for leaks and promptly repair them to minimize water wastage and associated pumping energy.
 - **Explore Rainwater Harvesting:** Although not currently implemented, given the presence of water reservoirs, re-evaluate the feasibility of a rainwater harvesting system to reduce reliance on underground water and associated pumping energy.

6.6 HVAC System Improvements

- **Observation:** The facility relies heavily on natural ventilation, but the presence of smoke and dust in certain areas and poor maintenance of existing cooling equipment (AC and fans) indicate a need for improvement.

- **Recommendation:**
 - **Improve Natural Ventilation:** Assess and optimize natural ventilation pathways to ensure adequate airflow and smoke dispersion. This might involve optimizing window openings or installing passive ventilation aids.
 - **Maintain Cooling Equipment:** Implement a regular maintenance schedule for the office AC unit, including cleaning or replacing filters, to ensure optimal performance and energy efficiency.
 - **Consider Exhaust Fans:** In areas with significant smoke or heat buildup, evaluate the feasibility of installing energy-efficient exhaust fans to improve air quality and worker comfort, especially if natural ventilation is insufficient.

6.7 Renewable Energy Integration

- **Observation:** A solar panel system is installed but not currently activated.
- **Recommendation:**
 - **Activate and Integrate Solar Panels:** Prioritize the activation and integration of the existing solar panel system into the facility's electrical supply. This will provide a clean, sustainable source of electricity, reduce reliance on grid power, and contribute to significant long-term operational cost savings. Conduct a technical assessment to ensure proper functioning and grid connection.

6.8 Operational and Maintenance Best Practices

- **Observation:** Safety signs are not visible, and the overall maintenance condition of some equipment is poor. There is no formal energy management system.
- **Recommendation:**
 - **Enhance Safety Signage:** Ensure all necessary safety signs are clearly visible throughout the facility, especially in areas with machinery and electrical hazards, to improve worker safety.
 - **Implement Preventive Maintenance Program:** Develop and strictly adhere to a comprehensive preventive maintenance program for all plant machinery and electrical systems. Regular maintenance not only extends equipment lifespan but also ensures optimal energy performance.
 - **Employee Training and Awareness:** Conduct regular training sessions for employees on energy efficiency best practices, the importance of power factor, and their role in energy conservation. Foster a culture of energy awareness throughout the organization.
 - **Waste Heat Recovery:** Investigate opportunities for waste heat recovery from processes like the recycling furnace or steam boiler, which could be utilized for pre-heating water or other processes, thereby reducing primary energy consumption.

7.0 Conclusion

The energy audit of Bismillah Foundry and Light Casting, Bright PET Washing Line Industries, has revealed significant opportunities for enhancing energy efficiency and reducing operational costs. The most critical area for immediate attention is the consistently low power factor, which currently results in substantial financial penalties. By addressing this issue through optimization or upgrade of the power factor correction system, the industry stands to achieve immediate and significant savings.

Furthermore, implementing a robust energy monitoring system, upgrading lighting infrastructure, optimizing motor and pump operations, improving HVAC system maintenance, and activating the existing solar panel system will collectively contribute to a more energy-efficient and sustainable operation.

References

- [1] L. Zhangjiagang Sevenstars Machinery co., "HDPE PET Bottle Flakes Washing Line," [Online]. Available: <https://www.ygplasticmachinery.com/hdpe-pet-bottle-flakes-washing-line/>.
- [2] "U.S. Department of Energy, Guide to Industrial Energy Efficiency, Washington," 2017. [Online]. Available: <https://www.energy.gov/sites/default/files/2017/12/f46/indutrial-energy-guide.pdf>.
- [3] R. H. a. T. H. M. Reuter, "Metals Recycling: Economic and Environmental Impacts, Delft," [Online].
- [4] D. K. A. a. L. A. R. H. Todd, "Manufacturing Processes Reference Guide," 1994.
- [5] K. G. S. a. J. D. Booker, "Manufacturing Process Selection Handbook," Butterworth-Heinemann, Oxford, UK, 2013.
- [6] "American Welding Society, Welding Handbook," AWS, Miami, FL, USA, 2004.
- [7] S. K. a. S. R. Schmid, "Manufacturing Engineering and Technology," Pearson, Boston, MA, USA, 2013.
- [8] M. P. Groover, "Fundamentals of Modern Manufacturing: Materials, Processes, and Systems," Wiley, Hoboken, NJ, USA, 2015.
- [9] "Molded Case Circuit Breakers - MCCBs," Schneider Electric, [Online]. Available: <https://www.se.com/in/en/product-subcategory/4230-molded-case-circuit-breakers-mccbs/>.
- [10] "PFR140/120/80/60 POWER FACTOR REGULATOR," MIKRO MSC BERHAD, [Online]. Available: <https://itmikro.com.my/pfr140-120-80-60-power-factor-regulator/>.
- [11] W. Z. P. D. L. (WZPDCL), "Annual Electricity bill of Bismillah Foundary Ltd from April 2024 to March 2025".
- [12] B. P. D. Board, "Tariff Rate: Effect from February 2024," [Online]. Available: https://bpdh.portal.gov.bd/sites/default/files/files/bpdh.portal.gov.bd/page/4a0f045a_1415_45f1_9e7c_02a8ed4f103c/2024-03-01-13-45-740f1fb3c50588f18077bdce93250b9a.pdf.
- [13] V. K. M. a. R. Mehta, Principles of Power System, New Delhi, India: S. Chand & Company Ltd, 2008.
- [14] I. E. S. o. N. America, The Lighting Handbook: Reference and Application, NY, USA, 2011.